Practical 4(LU Decomposition)

Question 1

In[•]:=
```
m = {{5, 2, 1}, {3, 7, 4}, {1, 1, 9}}
```

Out[•]=
```
{{5, 2, 1}, {3, 7, 4}, {1, 1, 9}}
```

In[•]:=
```
MatrixForm[m]
```

Out[•]//MatrixForm=

$$\begin{pmatrix} 5 & 2 & 1 \\ 3 & 7 & 4 \\ 1 & 1 & 9 \end{pmatrix}$$

In[•]:=
```
LUDecomposition[m]
```

Out[•]=

$$\left\{\left\{\{1, 1, 9\}, \{5, -3, -44\}, \left\{3, -\frac{4}{3}, -\frac{245}{3}\right\}\right\}, \{3, 1, 2\}, 0\right\}$$

In[•]:=
```
{lu, p, c} = LUDecomposition[m]
```

Out[•]=

$$\left\{\left\{\{1, 1, 9\}, \{5, -3, -44\}, \left\{3, -\frac{4}{3}, -\frac{245}{3}\right\}\right\}, \{3, 1, 2\}, 0\right\}$$

In[•]:=
```
{MatrixForm[lu], MatrixForm[p], c}
```

Out[•]=

$$\left\{\begin{pmatrix} 1 & 1 & 9 \\ 5 & -3 & -44 \\ 3 & -\frac{4}{3} & -\frac{245}{3} \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}, 0\right\}$$

In[•]:=
```
l = lu * SparseArray[{i_, j_} /; j < i → 1, {3, 3}] + IdentityMatrix[3]
```

In[•]:=

$$\left\{\{1, 0, 0\}, \{5, 1, 0\}, \left\{3, -\frac{4}{3}, 1\right\}\right\}$$

```
MatrixForm[l]
```

Out[•]=

$$\left\{\{1, 0, 0\}, \{5, 1, 0\}, \left\{3, -\frac{4}{3}, 1\right\}\right\}$$

Out[•]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 3 & -\frac{4}{3} & 1 \end{pmatrix}$$

In[ ]:=
```
u = lu * SparseArray[{i_, j_} /; j ≥ i → 1, {3, 3}]
```
Out[ ]=
```
SparseArray[ Specified elements: 6
             Dimensions: {3, 3} ]
```

In[ ]:=
```
MatrixForm[u]
```
Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 1 & 9 \\ 0 & -3 & -44 \\ 0 & 0 & -\frac{245}{3} \end{pmatrix}$$

In[ ]:=
```
l.u
```
Out[ ]=
```
{{1, 1, 9}, {5, 2, 1}, {3, 7, 4}}
```

In[ ]:=
```
MatrixForm[l.u]
```
Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 1 & 9 \\ 5 & 2 & 1 \\ 3 & 7 & 4 \end{pmatrix}$$

In[ ]:=
```
{m // MatrixForm, MatrixForm[l.u], p // MatrixForm}
```
Out[ ]=

$$\left\{\begin{pmatrix} 5 & 2 & 1 \\ 3 & 7 & 4 \\ 1 & 1 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 9 \\ 5 & 2 & 1 \\ 3 & 7 & 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}\right\}$$

In[ ]:=
```
mat = {{5, 2, 8}, {1, 2, 3}, {3, 4, 1}};
{lu, p, c} = LUDecomposition[mat]
```
Out[ ]=
```
{{{1, 2, 3}, {3, -2, -8}, {5, 4, 25}}, {2, 3, 1}, 0}
```

Question2

In[ ]:=
```
mat = {{5, 2, 8}, {1, 2, 3}, {3, 4, 1}};
MatrixForm[mat]
```
Out[ ]//MatrixForm=

$$\begin{pmatrix} 5 & 2 & 8 \\ 1 & 2 & 3 \\ 3 & 4 & 1 \end{pmatrix}$$

In[ ]:=
```
{lu, p, c} = LUDecomposition[mat]
```
Out[ ]=
```
{{{1, 2, 3}, {3, -2, -8}, {5, 4, 25}}, {2, 3, 1}, 0}
```

In[ ]:=
```
{MatrixForm[lu], MatrixForm[p], c}
```
Out[ ]=

$$\left\{\begin{pmatrix} 1 & 2 & 3 \\ 3 & -2 & -8 \\ 5 & 4 & 25 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, 0\right\}$$

In[ ]:=
```
l2 = lu * SparseArray[{i_, j_} /; j ≤ i → 1, {3, 3}];
```

In[ ]:=
```
l2 // MatrixForm
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 \\ 3 & -2 & 0 \\ 5 & 4 & 25 \end{pmatrix}$$

In[ ]:=
```
u2 = lu * SparseArray[{i_, j_} /; j > i → 1, {3, 3}] + IdentityMatrix[3];
u2 // MatrixForm
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -8 \\ 0 & 0 & 1 \end{pmatrix}$$

In[ ]:=
```
l2.u2 // MatrixForm
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 25 \\ 5 & 14 & 8 \end{pmatrix}$$

In[ ]:=
```
{mat // MatrixForm, MatrixForm[l2.u2], p // MatrixForm}
```

Out[ ]=

$$\left\{\begin{pmatrix} 5 & 2 & 8 \\ 1 & 2 & 3 \\ 3 & 4 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 25 \\ 5 & 14 & 8 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}\right\}$$

Question 3

In[ ]:=
```
Clear[All];
```

Clear: Symbol All is Protected.

In[ ]:=
```
A = {{5, 2, 1}, {3, 7, 4}, {1, 1, 9}}
```

Out[ ]=
```
{{5, 2, 1}, {3, 7, 4}, {1, 1, 9}}
```

In[ ]:=
```
b = {{10}, {21}, {12}}
```

Out[ ]=
```
{{10}, {21}, {12}}
```

In[ ]:=
```
{lu, p, c} = LUDecomposition[A];
l = lu * SparseArray[{i_, j_} /; j < i → 1, {3, 3}] + IdentityMatrix[3];
MatrixForm[l]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 3 & -\frac{4}{3} & 1 \end{pmatrix}$$

In[ ]:=
```
u = lu * SparseArray[{i_, j_} /; j ≥ i → 1, {3, 3}];
MatrixForm[u]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 1 & 9 \\ 0 & -3 & -44 \\ 0 & 0 & -\frac{245}{3} \end{pmatrix}$$

In[ ]:=
```
y = LinearSolve[l, b[[p]]];
x = LinearSolve[u, y]
```

Out[ ]=

{{1}, {2}, {1}}

In[ ]:=
```
A.x == b
```

Out[ ]=

True

Question4

In[ ]:=
```
A = {{2, 7, 5}, {6, 20, 10}, {4, 3, 0}};
b = {{0}, {4}, {1}};
{lu, p, c} = LUDecomposition[A];
l = lu * SparseArray[{i_, j_} /; j < i → 1, {3, 3}] + IdentityMatrix[3];
MatrixForm[l]
u = lu * SparseArray[{i_, j_} /; j ≥ i → 1, {3, 3}];
MatrixForm[u]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 11 & 1 \end{pmatrix}$$

Out[ ]//MatrixForm=

$$\begin{pmatrix} 2 & 7 & 5 \\ 0 & -1 & -5 \\ 0 & 0 & 45 \end{pmatrix}$$

In[ ]:=
```
y = LinearSolve[l, b[[p]]];
x = LinearSolve[u, y]
```

Out[ ]=

$$\left\{\left\{-\frac{1}{3}\right\}, \left\{\frac{7}{9}\right\}, \left\{-\frac{43}{45}\right\}\right\}$$

In[ ]:=
```
A.x
A.x == b
```

Out[ ]=

{{0}, {4}, {1}}

Out[ ]=

True